

THEORY AND METHODS ON SOME APPROACHES TO HAZARD RATE MODELLING

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Summary: Several types of hazard (failure) rate functions are considered while analyzing the probability of termination for the Poisson (renewal) process. Each event can lead to the process termination with a given, dependent on history probability. The black box hazard rate is defined as a conventional hazard (failure) rate of a continuous distribution function. The conditional hazard rate is defined by conditioning on information (history) at hand and the mean conditional hazard rate is the result of applying to the latter an operation of expectation. Some relationships between different types of hazard rate are considered. A simple reliability example is presented.

1. Introduction

The notion of the hazard (failure) rate is crucial in reliability and survival analysis. It is a very convenient tool for description of continuous lifetime random variables. If the hazard rate $\lambda_s(t)$ for some object, which will be called a system, exists, then the corresponding distribution function (d.f.) $F_s(t)$ can be defined by the standard exponential formula:

$$F_s(t) = 1 - \exp \left\{ - \int_0^t \lambda_s(u) du \right\}. \quad (1)$$

Why is the hazard rate $\lambda_s(t)$ so important and convenient in lifetime data analysis? One of the possible explanations is because it is already a conditional characteristic: $\lambda_s(t)dt$ can be interpreted as a probability of a failure in the infinitesimal interval $[t, t + dt)$ on condition that a system had survived (did not fail) in $[0, t)$. What, if some other information on the history of functioning of our system is available? This means that the process of functioning is observable (and not only the time of failure). The reasonable decision will be to use this information in defining the corresponding probability of failure in $[t, t + dt)$. Thus, our assessment of the probability of failure (in $[t, t + dt)$!) will be more adequate than in the case without additional information. This approach was actually formulated in Kalbfleisch and Prentice (1980) by means of the conditional hazard rate function. It is important to keep in mind that, as it was also stated in Kalbfleisch and Prentice (1980), a simple exponential representation (1) does not hold for the conditional hazard rate.

Can we still use the conditional hazard rate for obtaining $\bar{F}_s(t) \equiv 1 - F_s(t)$ – the probability of survival in the whole interval $[0, t)$, which is of the main interest in this paper? It will be shown that under some assumptions $\lambda_s(t)$ can be approximated by the mean conditional hazard rate, which is introduced in this paper and is defined as the mathematical expectation of the conditional hazard rate with respect to all possible realizations of the system's history (of functioning). The main advantage of this approach is that it is usually far easier to obtain the mean conditional hazard rate than $\lambda_s(t)$!

Consider an explanatory example of a system with 3 states. The system starts functioning at $t = 0$ in the state A and the transition to the state B occurs according to the absolutely continuous d.f. $F_{AB}(t)$ (with density $f_{AB}(t)$ and hazard rate $\lambda_{AB}(t)$). The transition from the state B to the absorbing state C (the failure) is governed by the d.f. $F_{BC}(t)$ ($f_{BC}(t)$, $\lambda_{BC}(t)$). There is no direct transition from state A to state C , though this case can also be easily considered. The information at hand (history) is the current state of the system and the corresponding duration of this state. Therefore the conditional hazard rates, describing transition to the absorbing state from the states A and B , are defined as

$$\lambda_{cA}(t) = 0 \text{ and } \lambda_{cB}(t) = \lambda_{BC}(t - t_B)du$$

respectively, where t_B denotes the random time of transition to the state B .

“Averaging the history”, we arrive at the following relation for the mean conditional hazard rate:

$$\lambda_{mc}(t) = 0 \cdot (1 - F_{AB}(t)) + \int_0^t f_{AB}(u) \lambda_{BC}(t - u) du = \int_0^t f_{AB}(u) \lambda_{BC}(t - u) du.$$

Finally, if the probability of system failure in $[0, t)$ is sufficiently small (see sections 3 and 4): $\lambda_s(t) \approx \lambda_{mc}(t)$.

This simple example describes the notions of conditional and mean conditional hazard rates. The hazard rate $\lambda_s(t)$ can also be directly derived in this case. In many practical situations, however, obtaining $\lambda_s(t)$ results in substantial difficulties while $\lambda_{mc}(t)$ can be obtained in a relatively simple way. This is a motivation of the study in sections 2 and 3.

The forthcoming can be considered as an illustration of the outlined approach within the framework of an example very important for applications. This example deals with a probabilistic description of a system subject to a

point process of shocks. Different shock models were thoroughly investigated in the literature (see Barlow and Proschan, 1975, Thompson, 1988, Hoyland and Rausand, 1994 to name a few).

2. Some definitions

Consider a system subject to a stochastic point process of shocks $\{N(t); t \geq 0\}$, where $N(t)$ is a random number of shocks in $[0, t)$; $N(a, b) \equiv N(b) - N(a)$, $b > a$. The point process can be described as in Cox and Isham (1980) by its complete intensity function:

$$\lambda(t; H_t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr \{N(t, t + \Delta t) = 1 \mid H_t\}}{\Delta t} \quad (2)$$

where H_t specifies the point process up to time t (history). For point processes without multiple occurrences $\lambda(t; H_t) dt$ can be interpreted as the probability of a shock occurrence in $[t, t + dt)$, $dt \rightarrow \infty$, given the process history up to t . The process history H_t (or part of it) will be regarded as information at hand for our models.

Definition (2) reduces to:

$$\lambda(t; H_t) = \lambda(t) \quad (3)$$

for the nonhomogeneous Poisson process (we shall omit the term "nonhomogeneous" in what follows) and to

$$\lambda(t; H_t) = \lambda(t')$$

for the renewal process, where t' is the time since the last event up to t , and $\lambda(x)$ is the hazard rate function of the underlying absolutely continuous distribution function $F(t)$, given by

$$F(t) = 1 - \exp \left\{ - \int_0^t \lambda(x) dx \right\}. \quad (4)$$

For the Poisson process relation (4) denotes the d.f. of the time of occurrence of the first event after the origin, and for the renewal process $F(t)$ is the d.f. of a cycle duration.

Let $h(t)$ denote the rate of occurrence or just the rate function of a point process. It is well known that for orderly point processes, assuming the limits exist:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr \{N(t, t + \Delta t) = 1\}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{E[N(t, t + \Delta t)]}{\Delta t}. \quad (5)$$

Thus, $h(t)dt$ can be interpreted as the probability of a shock occurrence in $[t, t + dt)$, $dt \rightarrow 0$. Let for the sake of definiteness an initial point be fixed at $t = 0$. It is clear that for the Poisson process $h(t) = \lambda(t)$ and that for the renewal process $h(t)$ becomes the renewal density function (Barlow and Proschan, 1975).

Assume that each shock (excluding the first one at $t = 0$) can cause a system's failure, terminating the point process. Under certain assumptions we shall be interested in deriving the d.f. of time to termination and analyzing conditional and unconditional probabilities of a system's failures in infinitesimal intervals of time. For this reason different types of hazard rate function will be considered and compared in the following section. Though the analysis is performed for the described setting, it can form the basis for certain generalizations.

3. Probability of termination for Poisson and renewal processes

Assume now that a shock, effecting an operating system at time $t \in (0, \infty)$ (thus formally excluding the first one at $t = 0$), independently of the previous shocks and its consequences to the system, causes a failure with probability $\theta(t)$ and does not cause any changes in the system with a complementary probability $1 - \theta(t)$. We shall consider this example of a terminating point process as a basic one for introducing different types of hazard rate functions.

An application of this setting to the safety at sea assessment was considered in Finkelstein (1998), where the shallows or other ships were the sources of external shocks effecting the given ship on a fixed route. Probability $\theta(t)$ was interpreted as a probability of an accident. Using the forthcoming results, this model can be generalized on the important case when the probability of an accident depends on the history of the point process of shocks (time since the last shock, for instance).

Let T_f denote the random time to failure (accident) and $F_f(t) = \Pr\{T_f \leq t\}$, the corresponding d.f. If $F_f(t)$ is absolutely continuous, then similar to (1):

$$P_f(t) = 1 - F_f(t) = \exp \left\{ - \int_0^t \lambda_f(x) dx \right\}, \quad (6)$$

where $P_f(t)$ is the corresponding survival function. Consider a Poisson process of shocks with rate $\lambda(t)$. Assume that $\theta(x)h(x)$ is integrable in $[0, t]$; $\forall t \in [0, \infty)$ and

$$\int_0^\infty \theta(x)\lambda(x)dx = \infty,$$

then the following simple formula takes place:

$$\lambda_f(t) = \theta(t)\lambda(t). \quad (7)$$

This result was first proved in Beichelt and Fischer (1980) (see also Block and Savits, 1985). For the time independent case $\theta(t) \equiv \theta$ the approach via equations (6)–(7) have been used by many authors in various situations; see, in particular, the method of “per demand hazard rate” in Thompson (1988).

We shall call $\lambda_f(t)$ defined by equation (6) a **black box hazard (failure) rate** of a system emphasizing the fact that it is the usual exponential representation for the absolutely continuous d.f. without considering any “internal” information. On the other hand, considering this internal information by means of the complete intensity function (2), we can define the relation for the **conditional hazard rate** $\lambda_c(t)$:

$$\lambda_c(t, H_t) dt = \Pr \{T_f \in [t, t + dt) | H_t, T_f(H_t) \geq t\} = \theta(t)\lambda(t, H_t)dt. \quad (8)$$

Condition $T_f(H_t) \geq t$ means that all shocks in $[0, t)$ were survived (for the specific configuration of shocks given by the history H_t). Conditional hazard rate is based on the internal “individual” information (Kalbfleisch and Prentice, 1980) and, as was stated in the introduction, bears no “usual” relationship with the corresponding survival function in the general case. At the same time it is clear that for the specific case of the Poisson process of shocks (8) turns to (7):

$$\lambda_c(t, H_t) = \lambda_f(t). \quad (9)$$

Hence, the above considerations actually present a new “non-technical” proof of (6)–(7). Indeed, if there is no dependency on the internal information in the relation for the conditional hazard rate, then the usual formula (6) for the (absolutely) continuous d.f. is valid.

It turns out that the conditional hazard rate can be dependent on the history H_t even for the Poisson process of shocks. Assume that the probability of failure depends on the history of the Poisson process: $\theta(t, H_t)$. This assumption is quite natural in applications because it states that the ability of a system to resist a shock depends on the past. The system, for instance, can “remember” the consequences of recent shocks and forget the consequences of distant shocks. For this model of dependency:

$$\lambda_c(t, H_t) dt = \Pr \{T_f \in [t, t + dt) \mid H_t, T_f(t) \geq t\} = \theta(t, H_t) \lambda(t) dt. \quad (10)$$

Based on equation (8) (or (9)), the **mean conditional hazard rate** $\lambda_{mc}(t)$ can be defined as

$$\lambda_{mc}(t) = E[\lambda_c(t), \{N(s); 0 \leq s < t\}], \quad (11)$$

where the expectation for the fixed $t \in [0, \infty)$ is taken with respect to all possible realizations of the point process in $[0, t)$. We shall discuss the general definition (11) while dealing with specific cases.

3.1 Poisson process. History: time since the last shock

Consider a specific case of the Poisson process, important for practical applications, when the probability of a failure depends only on z , the time elapsed since the previous shock: $\theta(t, H_t) = \theta(t, z)$. An application of this setting was treated in Finkelstein (1999), where the repairable system had to be operable at every instant of demand, occurring according to the Poisson process. Let $f(t)$ denote the probability density function (p.d.f.) for the governing d.f. (4). Then the black box hazard rate for the case under consideration is defined

in the usual way as:

$$\lambda_f(t) = \frac{f_f(t)}{\int_t^{\infty} f_f(u) du}, \quad (12)$$

where the corresponding p.d.f of time to failure is:

$$\begin{aligned} f_f(t) = & f(t)\theta(t) + f(t) \int_0^t \lambda(t_1)\bar{\theta}(t_1)\theta(t-t_1)dt_1 \\ & + f(t) \int_0^t \int_{t_1}^t \lambda(t_1)\bar{\theta}(t_1)\lambda(t_2)\bar{\theta}(t_2-t_1)\theta(t-t_2)dt_1dt_2 + \dots \end{aligned} \quad (13)$$

and for the sake of notation simplicity the first variable (current time) is omitted in $\theta(t, z)$.

It follows from (12) and (13) that even for this simplest case of dependence on history eventual formulas are complicated (in terms of computation) compared with surprisingly simple (6) and (7) for $\theta(t, H_t) = \theta(t)$.

Thus, obtaining simple approximations of $\lambda_f(t)$ can be **very important in applications**. One of the possible ways to do it is via the mean conditional rate (11) and it will be shown in this section. On the other hand, apart from approximation, obtaining $\lambda_{m,c}(t)$ can be important in itself in reliability and survival analysis applications. The reason is that this characteristic defines some averaged hazard, which is of interest while describing and comparing probabilistic behaviour of different objects.

Let time since the previous shock z , which constitutes H_t , be fixed. Then:

$$\lambda_c(t, z) = \theta(z)\lambda(t). \quad (14)$$

Multivariate outliers in the regressor space are often referred to as leverage observations (Mason and Gunst, 1985). Chatterjee and Hadi (1988) remark that high-leverage observations tend to influence the eigenstructure of the regressor matrix X . They refer to observations that create or obscure near collinearities as collinearity-influential observations. Therefore a small condition index (or variance inflation factor (VIF)) does not necessarily mean that collinearity is absent. On the other hand, one or two observations may be responsible for a large condition index (or VIF). We refer to observations that create collinearity as type A observations and to observations that obscure collinearity as type B (see Figure 1). Type A observations inflate variances, covariances and correlations, resulting in large condition indices and VIFs. The converse is true for type B observations. Type B observations sustain spurious dimensions which may lead to non-significant regressors being included in the regression model. This is a potential danger in automated regression procedures. Note that while collinearity-influential observations usually have high leverage, not all high leverage observations are collinearity-influential (Chatterjee and Hadi, 1988). These observations, a combination of type A and type B observations, are labelled type AB observations. They have little or no effect on the correlation structure of the regressors.

Multivariate outliers have an interrelated effect on the estimation of location, scale (i.e. variance) and orientation (i.e. correlation). Multivariate dispersion is composed of the latter two components, namely scale and orientation (Gnanadesikan and Kettenring, 1972). Collinearity-influential observations are primarily involved with these facets of multivariate data. We illustrate how the factors of a decomposition of the VIF relate to these

to u gives the following function:

$$\lambda(t_1, t_2, \dots, t_k, u) = \begin{cases} \lambda(u)\theta(u), & 0 \leq u < t_1 \\ \lambda(u)\theta(u - t_1), & t_1 \leq u < t_2 \\ \dots \\ \lambda(u)\theta(u - t_k), & t_k \leq u < t \end{cases} \quad (18)$$

and the conditional survival function is defined by

$$P_f(t_1, t_2, \dots, t_k, t) = \exp \left\{ - \int_0^t \lambda(t_1, t_2, \dots, t_k, u) du \right\}. \quad (19)$$

Thus $\lambda(t_1, t_2, \dots, t_k, u) du$ can be interpreted as probability of the system failure in $[u, u + du)$ on condition that there were no failures before for the given (prior to the failure) history of the Poisson process in $[0, t)$. Thus, due to the definition of the conditional hazard rate (10), for the specific case under consideration:

$$\lambda_c(t, H_t) = \lambda(t_1, t_2, \dots, t_k, u).$$

It should be noted that equation (16) can also be obtained via (18).

For the "sufficiently reliable systems" when $1 - P_f(t) \ll 1$ ($\lambda(z) \rightarrow 0, 0 \leq z < t$, for instance, or $\theta(z) \rightarrow 0, 0 < z \leq t$) only two terms in the corresponding Taylor's expansion in a series for (17) can be considered:

$$\begin{aligned} P_f(t) &\approx \left[1 - E \left[\int_0^t \lambda_u du \right] \right] = 1 - \int_0^t \lambda_{m,c}(u) du \\ &\approx \exp \left\{ - \int_0^t \lambda_{m,c}(u) du \right\} \end{aligned} \quad (20)$$

This means that under the same assumption the following asymptotic equation holds:

$$\lambda_f(t) = \lambda_{m,c}(t)(1 + o(1)). \quad (21)$$

Thus, $\lambda_{m,c}(t)$ is a simple approximation for $\lambda_f(t)$ and this fact can be very important in applications.

3.2 Renewal process. History: time since the last shock

For the renewal process relation (10) can be modified as:

$$\lambda_c(t, H_t) dt = \Pr \{T_f \in [t, t + dt) \mid H_t, T_f(t) \geq t\} = \theta(t, H_t) \lambda(z) dt$$

and (18) will turn to

$$\lambda(t_1, t_2, \dots, t_k, u) = \begin{cases} \lambda(u)\theta(u), & 0 \leq u < t_1 \\ \lambda(u-t_1)\theta(u-t_1), & t_1 \leq u < t_2 \\ \dots \\ \lambda(u-t_k)\theta(u-t_k), & t_k \leq u < t \end{cases}$$

which, similar to (16), leads to the following relation for the mean conditional hazard rate:

$$\lambda_{m,c}(t) = \theta(t)f(t) + \int_0^t h(t-z)\theta(z)f(z)dz \quad (22)$$

Similar to (13):

$$\begin{aligned} f_f(t) &= f(t)\theta(t) + \int_0^t f(t_1)\bar{\theta}(t_1)f(t-t_1)\theta(t-t_1)dt_1 \\ &+ \int_0^t \int_{t_1}^t f(t_1)\bar{\theta}(t_1)f(t_2-t_1)\bar{\theta}(t_2-t_1)f(t-t_2)\theta(t-t_2)dt_1dt_2 + \dots \end{aligned} \quad (23)$$

and the corresponding relation for $\lambda_f(t)$ is also based on the infinite series.

For the specific case $\theta(z) = \theta$, (22) reduces to

$$\lambda_{m,c}(t) = \theta f(t) + \int_0^t h(t-z)\theta f(z)dz = \theta h(t). \quad (24)$$

Similar to (20), we can conclude that for $\theta(z) \rightarrow 0$, $0 < z \leq t$ in the time-dependent case (23) or for $\theta \rightarrow 0$ in the specific case (24), the mean conditional hazard rate $\lambda_{m,c}(t)$ defines asymptotic approximation for the black box failure rate $\lambda_f(t)$.

The described approach can be easily generalized on the "more distant" time dependency in $\theta : \theta(z_1, z_2), \theta(z_1, z_2, z_3) \dots$, where z_1 is the time since the last shock, z_2 is the time since the last but one shock etc. It should be understood that this generalization will lead to additional terms in equations (16) and (22).

4. Example

We shall illustrate different types of hazard rate function using a simple reliability setting specifying an illustrative example of section 1. Consider a system of two identical components with exponential failure time distributions:

$F(t) = 1 - \exp\{-\lambda t\}$. The first component starts functioning at $t = 0$. When it fails, the second component is instantaneously switched into operation. The system fails when the second component fails. Therefore the d.f. of time to the system's failure is a convolution of two i.i.d. exponential random variables:

$$F_f(t) = 1 - \exp\{-\lambda t\} [1 + \lambda t]. \quad (25)$$

The black box failure rate can be obtained directly

$$\lambda_f(t) = \frac{\lambda^2 t}{1 + \lambda t}. \quad (26)$$

The information at hand at some moment of time $u \in [0, t)$ for the operating system is just the number of operable components. If both components are operable at u then the conditional hazard rate is 0 at this moment of time

(and it is equal to λ , if only one component is operable while the second had already failed):

$$\lambda_c(u, H_u = 2) = 0; \lambda_c(u, H_u = 1) = \lambda. \quad (27)$$

The first equality follows from (26) after substituting $t = 0$. For other instants of time relation (26) gives results different from (27). Thus, the availability of information is very important!

The hazard rate process $\{\lambda_t, t \geq 0\}$ is defined as the process equal to 0 in $[0, u)$ and equal to λ in $[u, \infty)$ while variable u is exponentially distributed with parameter λ . The mean conditional hazard rate is defined as in (11):

$$\lambda_{m,c}(t) = E[\lambda_u, 0 \leq u < t] = \lambda(\lambda t \exp\{-\lambda t\}) = \lambda^2 t \exp\{-\lambda t\}. \quad (28)$$

The survival function for this elementary setting can also be defined according to (17) and this leads to the same result as in (25):

$$\begin{aligned} P_f(t) &= E \left[\exp \left\{ - \int_0^t \lambda_u du \right\} \right] = \exp\{-\lambda t\} \\ &\quad + \lambda \int_0^t \exp\{-\lambda u\} \exp\{-\lambda(t-u)\} du \\ &= \exp\{-\lambda t\} [1 + \lambda t]. \end{aligned}$$

We can compare $\lambda_f(t)$ and $\lambda_{m,c}(t)$ in this example. It is evident that asymptotic relation (21) holds for λt sufficiently small. Hence, $\lambda_{m,c}(t)$ is a good approximation for $\lambda_f(t)$ for sufficiently reliable systems. This fact is not so important in this example, because a simple formula for $\lambda_f(t)$ is available, but in other situations it can be very helpful.

Termination of the process can easily be considered for this example. Assume that in addition to the described setting the failure of the first component can result in the failure of the system with probability θ and in switching on the second component with probability $1 - \theta$. A possible practical interpretation is that the switching device is not ideal (as was assumed), but can fail with probability θ when addressed. Equation (25) will turn to

$$F_f(t) = 1 - \exp \{-\lambda t\} [1 + (1 - \theta) \lambda t]$$

and all subsequent formulas can easily be rewritten. The generalization on the possible dependence on u in probability of termination is straightforward. The systems of more than two components can be considered in a similar way.

5. Concluding remarks

We have considered the following types of hazard rate functions:

- (a) The black box hazard rate.
- (b) The conditional hazard rate.
- (c) The mean conditional hazard rate.
- (d) The hazard rate process.

The black box hazard rate $\lambda_f(t)$ is commonly used for characterizing absolutely continuous lifetime distribution functions. In various situations obtaining $\lambda_f(t)$ results in cumbersome or implicit formulas. The conditional hazard rate, based on the individual information, is also very important and has a good natural meaning. It can be interpreted as the probability of a failure in $[t, t + dt)$ on condition that the system did not fail before given the history of the process in $[0, t)$. If this information is available, then our assessment of

the probability of failure in $[t, t + dt)$ is much better than the one based on $\lambda_f(t)$.

The mean conditional hazard rate $\lambda_{m,c}(t)$ is a new notion. It is obtained in a natural way by relation (11), which defines the mean of the corresponding hazard rate process. As it was stated, $\lambda_{m,c}(t)$ is a good approximation for $\lambda_f(t)$ for the sufficiently reliable systems, and this can be very helpful in various applications.

Different types of hazard rate functions were discussed in this paper within the framework of the concrete application. Obtaining the probability of termination for the Poisson and renewal processes is an important applied problem. The interpretation of safety at sea of this problem was pointed out.

There are some other evident applications of the developed approach (see, for instance, Finkelstein, 1999).

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References

- BARLOW, R., FROSHAN, F. (1975). *Statistical theory of reliability and life testing. Probability Models*. Holt, Rinehart & Winston.
- BEICHELT, F., FISCHER, K. (1980). General failure model applied to preventive maintenance policies. *IEEE Trans. Reliability*, **29**, 39–41.
- BLOCK, H.W., BORGES, W., SAVITS, T.H. (1985). Age dependent minimal repair. *J. Appl. Prob.* **22**, 370–386.

- COX, D.R., ISHAM, V. (1980). *Point processes*. Chapman & Hall.
- FINKELSTEIN, M.S. (1998). A point process stochastic model with application to safety at sea. *Reliability Engineering and System Safety*, **60**, 227–223.
- FINKELSTEIN, M.S., (1999). Multiple availability on stochastic demand. *IEEE Trans. Reliability*, **48**, 19–25.
- HOYLAND, A., RAUSAND, M. (1994). *System reliability theory. Models and statistical methods*. John Wiley & Sons.
- KALBFLEISCH, J.D., PRENTICE, R.L. (1980). *The statistical analyses of failure time data*. John Wiley & Sons.
- KEBIR, Y. (1991). On hazard rate processes. *Naval Research Logistics* **38**, 865–876.
- THOMPSON, W.A., Jr. (1988). *Point process models with applications to safety and reliability*. Chapman & Hall.